

ESSENTIAL SKILL

Water

Find It. Purify It. Protect It.

Water is one of the first things we must think about in an emergency.

- How much water you need.
- Where you can find it.
- How to make questionable water safe.
- How to store clean water without contaminating it.

IMMEDIATE ACTION

If you need water right now

1

Use stored or bottled drinking water first.

2

If the water may contain germs, boil it.

3

If you cannot boil it, use an approved disinfection method.

4

If you suspect fuel, chemicals, or other toxic contamination, find another source.

IMPORTANT

Boiling kills germs. It does not remove most chemicals.

PLANNING AMOUNT

How much water?

A good emergency starting point is:

1 gallon

PER PERSON, PER DAY

This covers drinking and some basic cooking and hygiene needs. Keep more when possible for:

- Hot weather, illness, or heavy physical activity.
- Children and older adults.
- Pets.

For home preparedness, storing several days is good. A two-week supply is even better when practical.

START WITH THE SAFEST

Where can you find water?

Possible sources include:

- Stored drinking water.
- Bottled water.
- A safe public water supply.
- A safe well.
- Household plumbing.
- An appropriate household water heater.
- Rainwater.
- Springs.
- Streams and rivers.
- Lakes and ponds.

Outdoor water should normally be treated before drinking. Clear water is not necessarily safe water.

How to boil water

Boiling is the simplest dependable method for killing most disease-causing germs.

STEP 1 If the water is cloudy, let sediment settle or pass the water through a clean cloth or coffee filter.

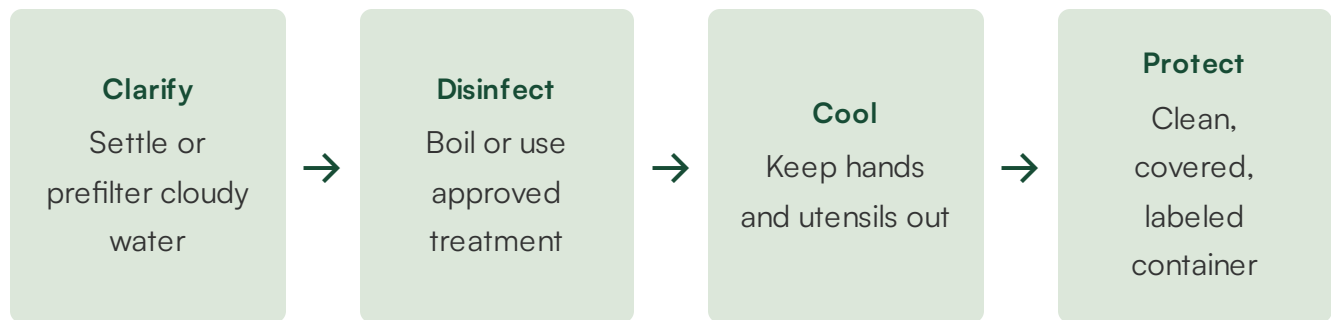
STEP 2 Bring the water to a rolling boil.

STEP 3 Boil for 1 minute. At elevations above approximately 6,500 feet, boil for 3 minutes.

STEP 4 Allow the water to cool naturally.

STEP 5 Put it into a clean, covered container.

SAFE-WATER SEQUENCE



EMERGENCY BACKUP

If you cannot boil

Ordinary unscented household chlorine bleach can be an emergency backup. Check the label first.

For clear water and bleach containing 5%—9% sodium hypochlorite:

1 gallon + 8 drops

1. Stir well.
2. Wait at least 30 minutes.
3. Keep the treated water covered.

For cloudy, murky, colored, or very cold water, CDC guidance calls for double the amount.

NEVER USE

- Scented bleach.
- Color-safe bleach.
- Bleach mixed with cleaners.
- Products when you do not know what they contain.

Never guess. Check the label and current public-health instructions.

KNOW YOUR EQUIPMENT

What about water filters?

Filters can be extremely useful. But remember:

A filter is not automatically a purifier.

Different filters remove different things. Some remove parasites. Some remove bacteria. Many do not reliably remove viruses.

Before depending on a filter, know:

- What it removes and what it does not remove.
- How often it must be cleaned or replaced.
- Whether the water also needs to be disinfected.

Follow the manufacturer's instructions.

FIND ANOTHER SOURCE

When water cannot be made safe

Do not assume every contaminated water source can be repaired by boiling or filtering.

Water suspected of containing gasoline, fuel oil, pesticides, toxic chemicals, or industrial contamination should not simply be boiled and consumed.

Find another water source.

RECOMMENDED SUPPLIES

Simple water kit

☐ Stored drinking water

☐ Food-grade water containers

☐ Large cooking pot

☐ Safe heat source

☐ Clean cloth or coffee filters

☐ Portable water filter

☐ Unscented household bleach

☐ Medicine dropper

☐ Water purification tablets

☐ Buckets

☐ Funnel

☐ Labels and permanent marker

PREVENT RECONTAMINATION

The two-container rule

This one habit can prevent many problems. Keep two containers clearly marked:

CONTAINER 1

Untreated water

CONTAINER 2

Safe drinking water

Never put purified water back into a dirty container. Do not dip dirty hands, cups, or utensils into stored drinking water. A covered container with a pouring spout is best.

AVOID THESE

Common mistakes

✗ Assume clear water is safe.

✗ Drink directly from a stream because it looks clean.

✗ Put clean water into a dirty container.

✗ Guess at bleach amounts.

✗ Assume every camping filter removes every danger.

✗ Waste drinking water when untreated water could serve another purpose.

✗ Assume boiling removes chemicals.

TRY THIS NOW

Water practice

Take one gallon of water and pretend the normal water supply has stopped.

1. Find a way to filter visible sediment.
2. Boil the water.
3. Cool it.
4. Put it into a clean container.
5. Label it **Safe Water**.

Could I still do this if the electricity and running water were both off?

If not, you have discovered what you need to prepare.

BEFORE AN EMERGENCY

Quick checklist

☐ I know how much drinking water I have.

☐ I know where additional water could come from.

☐ I can boil water without grid electricity.

☐ I have clean water containers.

☐ I know how my water filter works.

☐ I have a backup purification method.

☐ I can keep untreated water separate from safe water.

REMEMBER

Water does not have to look dirty to be dangerous.

Boiling is usually the simplest way to kill germs.

Chemical contamination requires another source.

FIND IT. PURIFY IT. PROTECT IT.

Official guidance

Emergency instructions can change. Check current local public-health advice whenever it is available.

CDC: How to Make Water Safe in an Emergency(<https://www.cdc.gov/water-emergency/about/index.html>)

CDC: Create and Store an Emergency Water Supply (<https://www.cdc.gov/water-emergency/about/how-to-create-and-store-an-emergency-water-supply.html>)

Ready.gov: Water(<https://www.ready.gov/water>)
